



1.3 Fetal Physiology

1. Placenta platform

→ Fetal circulation = placenta → umbilical vein (oxygenated) → DUCTUS VENOSUS (bypasses fetal liver) → IVC → R atrium → [(1) FORAMEN OVALE (bypasses fetal lungs) → L atrium → L ventricle <OR> (2) R ventricle → pulmonary arteries → DUCTUS ARTERIOSUS (bypasses fetal lungs)] → aorta → umbilical arteries (deoxygenated) → placenta

2. Umbilical vein arch

→ Fetal circulation = placenta → umbilical vein (oxygenated) → DUCTUS VENOSUS (bypasses fetal liver) → IVC → R atrium → [(1) FORAMEN OVALE (bypasses fetal lungs) → L atrium → L ventricle <OR> (2) R ventricle → pulmonary arteries → DUCTUS ARTERIOSUS (bypasses fetal lungs)] → aorta → umbilical arteries (deoxygenated) → placenta

3. 2 umbilical artery ropes

→ Fetal circulation = placenta → umbilical vein (oxygenated) → DUCTUS VENOSUS (bypasses fetal liver) → IVC → R atrium → [(1) FORAMEN OVALE (bypasses fetal lungs) → L atrium → L ventricle <OR> (2) R ventricle → pulmonary arteries → DUCTUS ARTERIOSUS (bypasses fetal lungs)] → aorta → umbilical arteries (deoxygenated) → placenta

4. Black arrow path + “to bypass” sign on liver slide

→ Fetal circulation = placenta → umbilical vein (oxygenated) → DUCTUS VENOSUS (bypasses fetal liver) → IVC → R atrium → [(1) FORAMEN OVALE (bypasses fetal lungs) → L atrium →

L ventricle <OR> (2) R ventricle → pulmonary arteries → DUCTUS ARTERIOSUS (bypasses

fetal lungs)] → aorta → umbilical arteries (deoxygenated) → placenta

5. Fallen VWater Duct on liver slide

→ Fetal circulation = placenta → umbilical vein (oxygenated) → DUCTUS VENOSUS (bypasses

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L ventricle <OR> (2) R ventricle → pulmonary arteries → DUCTUS ARTERIOSUS (bypasses

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6. R atrium kid with crossed out lungs shirt throwing frisbee through oval hole to L atrium kid

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fetal lungs)] → aorta → umbilical arteries (deoxygenated) → placenta

7. Green arrow path

→ Fetal circulation = placenta → umbilical vein (oxygenated) → DUCTUS VENOSUS (bypasses

fetal liver) → IVC → R atrium → [(1) FORAMEN OVALE (bypasses fetal lungs) → L atrium →

L ventricle <OR> (2) R ventricle → pulmonary arteries → DUCTUS ARTERIOSUS (bypasses

fetal lungs)] → aorta → umbilical arteries (deoxygenated) → placenta

8. Fallen air duct

→ Fetal circulation = placenta → umbilical vein (oxygenated) → DUCTUS VENOSUS (bypasses

fetal liver) → IVC → R atrium → [(1) FORAMEN OVALE (bypasses fetal lungs) → L atrium →

L ventricle <OR> (2) R ventricle → pulmonary arteries → DUCTUS ARTERIOSUS (bypasses

fetal lungs)] → aorta → umbilical arteries (deoxygenated) → placenta

9. Proslugger bat + “E” jersey

→ ductus arteriosus kept open by prostaglandin E (NSAIDs can cause premature closure)

10. Detergent cleaning lungs

→ fetal lung maturation = needs surfactant (phospholipids, lecithin) to keep alveoli open, if premature → give corticosteroids (betamethasone)

11. Moon on detergent bucket

→ fetal lung maturation = needs surfactant (phospholipids, lecithin) to keep alveoli open, if premature → give corticosteroids (betamethasone)

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fetal hemoglobin = HgbF ($2\alpha+2\gamma$ chains) = higher affinity for O₂

13. $2\alpha+2\gamma$ balls

→ fetal hemoglobin = HgbF ($2\alpha+2\gamma$ chains) = higher affinity for O₂

14. Baby drinking yellow drink + peeing

→ fetal urine = amniotic fluid

15. Fan with bowtie and “2” jersey

→ fetal thyroid starts functioning during 2nd trimester (prior, maternal T₃/T₄ crosses the placenta)

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“K” syringe cone next to liver structure